

CIS 635 Knowledge Discovery & Data Mining

Introduction to ML

ML introduction

What is ML? Anyone wants to share your thoughts??

ML introduction

What *is* Machine Learning?

Machine Learning (ML) is when a computer learns patterns from data and improves its performance **without being explicitly programmed** for every task.

What *is NOT* Machine Learning?

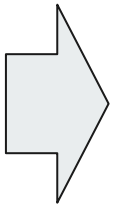
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Example Applications



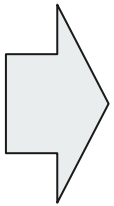
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   - Follows strict math rules coded by a programmer.
   - It doesn't "learn" how to calculate—it just executes instructions.
1. Digital Alarm Clock:
   - Goes off at a set time programmed by the user.
   - No learning—just performs based on what you set.
1. Static Website:
   - Displays fixed content to all users.
   - Doesn't adapt or learn from user behavior.
1. Microwave Oven Timer:
   - Follows direct input commands (e.g., heat for 2 minutes).
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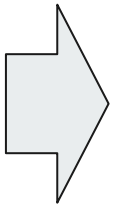
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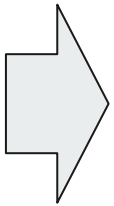
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   - Over time, it gets better at predicting which emails are spam—even if it's never seen that exact message before.
2. Netflix Recommendations:
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3. Voice Assistants (e.g.…
   - Learn your …
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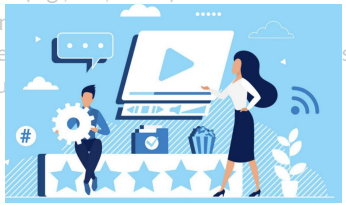
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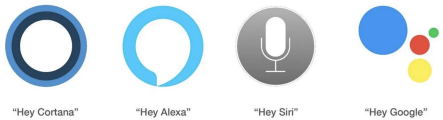
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   - Understand different accents or phrases better the more you use them.

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5. Conversational Age
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    - Improve



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5. **Conversational Agent (ChatGPT)**
    - Learns from massive text data
    - Improves using human feedback.



Let's Check with our Understanding!

Q. What was the average daytime temperature in Grand Rapids in August 2024?

Is it a ML problem?

Let's Check with our Understanding!

Q. What was the average daytime temperature in Grand Rapids in August 2024?

Clue: This information is already known.

Is it a ML problem?

Let's Check with our Understanding!

Q. What will be the average daytime temperature in Grand Rapids in August 2027?

Is it a ML problem?

Let's Check with our Understanding!

Q. What will be the average daytime temperature in Grand Rapids in August 2026?

Is it a ML problem?

Clue: This information is unknown and involves uncertainty.

QA